

HW_01

Henry Traynor

Due: 25 August

Assignment

Create a self-contained R script (or better yet an Rmarkdown file) to do the following:

Simulate logistic regression data according to $y_i \sim \text{Bern}(p_i)$ and $\text{logit}(p_i) = \beta_0 + \beta_1 x_i$ with $\beta_0 = -1$ and $\beta_1 = 1$. Generate the covariate using `x <- rnorm(n=100)`.

```
beta_0 <- -1
beta_1 <- 1
x <- rnorm(n=100)

y <- rbinom(100,
            size=1,
            prob = plogis(beta_0 + beta_1 * x))
```

Fit a logistic regression model to the simulated data using the `glm` function. Create a figure showing x and y with the fitted regression line. Use `predict` to get the fitted line.

```
library(ggplot2)
logistic_regression <- glm(y ~ x,
                           family = binomial)
predicted_data <- data.frame(
  x = seq(min(x), max(x), length.out = 100)
)

predicted_data$y_predicted <- predict(logistic_regression,
                                     newdata = predicted_data,
                                     type = "response")

data <- data.frame(x, y)

ggplot() +
  geom_point(
    data = data,
    aes(x = x, y = y, color = "Observed data")
  ) +
  geom_line(
    data = predicted_data,
    aes(x = x, y = y_predicted, color = "Fitted regression")
  ) +
  labs(
    x = "x",
```

```
y = "y",  
color = "Data"  
)
```

